



Research on dynamics model of the motion clamping mechanism in injection molding machines addressing inertia and deformation

Zhiqiang He^a, Fugang Zhai^{a,*}, Yin Zhang^{a,b}, Weibo Wang^a, Dong Li^a, Ningdi Zhu^c, Jianbo Ruan^c

^a School College of Mechanical Engineering, Yanshan University, Qinhuangdao 066004, China

^b Department of mechanical and electrical engineering, Qinhuangdao Vocational and Technical College, Qinhuangdao 066100, China

^c Haitian Plastics Machinery Group, Ningbo 315801, China

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ABSTRACT

Motion clamping mechanisms (MCM) play a crucial role in scenarios requiring multi-phase motions, notably in processes like the injection molding machine. The working process of mechanicals, which necessitates precise positioning, swift response, and the seamless coordination of multi-phase movements, is primarily impeded by the inertia and deformation of the components. This paper introduces modelling techniques for the four motion phases of the MCM. The methods include the symmetric recursion method based on Newton-Euler dynamics and the deformation distribution method based on the ratio of component stiffness coefficients. By considering inertia and deformation, models are established for each sub-phase of the motion process, and tackles complex friction challenges associated with multiple friction pairs, as well as resolving the issue of fixed joint space mechanisms being unable to be unilaterally deduced. A logic based on dual parameters of position and pressure, is devised to create a comprehensive dynamic model of the mechanism's entire process. Experimental validation demonstrates that the accuracy of sub-model exceeds 90 %, while the continuous model maintains an accuracy of over 97 % in multiple tests. The method improves the accuracy of identification and switching by 16.7 %, resulting in a high-precision dynamic describe of the entire process of the MCM, thereby validating its effectiveness.

1. Introduction

Motion clamping mechanisms (MCM) play a crucial role in various industrial applications, notably in equipment such as presses and injection molding machines (IMM) [1,2]. These mechanisms undergo a multi-phase working process encompassing moving contact, elastic loading, and workpiece protection, which necessitates precise positioning, swift response, and the seamless coordination of multi-phase movements. These requirements pose significant challenges for high-performance control of the equipment [3], with crucial influencing factors being the inertia and deformation of the components [4,5]. However, in the face of complex transmission mechanisms, these two factors are often overlooked or simplified. Therefore, proposing a modelling approach that comprehensively considers both inertia and deformation is of crucial importance for enhancing the control performance and production efficiency of the injection molding process.

In the injection molding process, the rapid movement of the moving platen of the MCM ensures a close fit with the mould. By utilizing the

elastic deformation of the connecting rod, the mould clamping is achieved. This motion process is a critical step in injection molding production, determining production efficiency and performance [6]. The research on the MCM has focused on aspects such as structural and motion optimization, as well as stability adjustment. Huang, Le et al. established the dynamic model of the MCM using the virtual displacement principle and investigated the influence of key design parameters on the performance of the MCM [7,8]. Hsu et al. utilized the inverse dynamics algorithm to obtain the input absolute electrical energy of the mechanism, enabling energy-efficient trajectory planning [9]. Zhao et al. conducted mechanical derivations of the MCM, disregarding the influence of gravity and inertia tensor, and constructed the velocity model of the mechanism for stability management [6]. The research on key components and performance indicators concentrates on the analysis and optimization of tie bars [10–13] and monitoring and controlling the clamping force [14–17]. This segment of the research relies on the test results and lacks theoretical support. In the identification and switching between phases, several scholars have explored functional

* Corresponding author.

E-mail address: zhafugang@ysu.edu.cn (F. Zhai).

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models such as genetic algorithm [18], fuzzy algorithm [19], and load model [20]. Furthermore, there have been innovations in the clamping methods of the mechanism [1,21,22].

Undeniably, the aforementioned research has made significant contributions to performance enhancement of the MCM. However, the research process has overlooked factors such as inertia, deformation, and friction, which have substantial impacts on dynamics. The excessive simplification of real-world scenarios can lead to limitations in research methods and conclusions, especially in high-speed and large-inertia conditions such as those found in IMMs. In light of the deficiencies in current research on the MCM, the specific influences and requirements can be summarized as follows:

- In the platen moving phase of the MCM, the theoretical foundation relies on the principle of virtual displacement. The content completely overlooks the impact of inertia and friction on moving components [9], failing to provide a comprehensive representation of the force-position relationship [23]. This is also the root cause of the current errors in control effectiveness [20]. To address this limitation, it becomes imperative to introduce a modelling theory that integrates inertia, enhancing the comprehensiveness and transparency of the representation of state information for the MCM.
- In the clamping phase of the MCM, the platen relies on the elastic deformation of components to secure the mould. This process entails the elastic deformation of multiple components, each composed of different materials and assuming diverse postures. Simply force analysis proves challenging in establishing the accurate mapping relationship between the platen position and the clamping force [18]. There is a compelling need to introduce a modelling method grounded in the elastic deformation of multiple components. Such an approach is essential to accurately characterize the clamping force during the clamping phase.
- The motion process of the MCM involves multiple phases of alternating cycles. Currently, the control switching between phases hinges on high-frequency response and high-gain open-loop control utilizing position signals. This approach frequently leads to challenges like premature deceleration in feedforward control and impacts on the mould due to lag control. These issues significantly impact production efficiency and may pose potential safety hazards [20,21]. The effectiveness of existing research has not been validated in practical applications. Therefore, there is an urgent need for more effective identification criteria and switching logic to achieve active control and precise switching of multi-phase motions.

In addition, in practical applications, there are also some issues, such as high dependence on parameter tuning experience for the MCM [24] and low levels of equipment digitization, making it difficult to integrate emerging technologies [25]. In contrast to the injection system of the IMM, which has undergone extensive intelligent research on equipment performance optimization [26,27] and process matching [28–30], the evolution of the MCM lags noticeably behind. This deficiency poses a significant obstacle to the performance of overall machine and intelligent processes.

In light of the challenges and needs outlined in the MCM, this paper proposes new modelling methods addressing both inertia and deformation. It further designs a logic for phase switching based on measurable parameters, creating a comprehensive dynamic model for the entire process. The proposed methods and models are validated on an IMM testbed, demonstrating their effectiveness. The specific innovations and contributions are as follows:

- (1) Unlike the model oversimplification in current theoretical methods, the proposed symmetrical recursion method (SRM) based on Newton-Euler dynamics, when combined with the characteristics of mechanical transmission, effectively accounts for the impact of component friction, inertia, and coupled inertia

on dynamics. This method enables precise description of the dynamics during the platen moving phase, enhancing the accuracy of the dynamic model and contributing to improving the control precision of the injection molding process.

- (2) Addressing the deformation effects of the components, a deformation distribution method (DDM) is introduced based on the ratio of component stiffness coefficients. Leveraging the insight that greater component stiffness hinders elastic deformation, a mapping relationship between the position of the cylinder and the deformation of components is established. This facilitates high-precision modelling of the clamping force during the clamping phase, which is advantageous for the precise application of mould clamping force.
- (3) Compared to the current single-parameter recognition method, which solely considers position, the multi-phase identification criteria and switching logic are designed based on both displacement and pressure. This logic links the dynamics models of the platen moving and clamping phase, improves the reliability and smoothness of phase identification in the injection molding process. This improvement contributes to enhancing operational efficiency.

The rest of this article is organized as follows: [Section 2](#) introduces the multi-phase motions of the MCM and analyses the kinematics of the mechanism. [Section 3](#) provides a detailed introduction to the dynamic modelling methods. [Section 4](#) conducts experiment on the method and analyses the results. Finally, [Section 5](#) summarizes and discusses the innovations of this article.

2. System description and kinematic analysis

2.1. System description of MCM

The MCM is shown in [Fig. 1](#). The platen is guided by the tie bar and, driven by the clamping hydraulic cylinder, the movement is transmitted through the connecting rod to complete the reciprocating clamping motion.

The work process of the MCM includes two processes: mould closing and mould opening, as shown in [Fig. 2](#). The mould closing process consists of two phases: platen moving and clamping mould. Driven by the clamping cylinder, the platen moves until the mould is fitted, completing the mould moving phase. Then, the cylinder extends, pressing the platen. The deformation force generated by lengthening of the tie bar and the self-locking characteristics of the MCM are utilized to clamp the mould, completing the clamping phase.

After the injection-moulded product has cooled and formed, the mould opening process is carried out, including two phases: opening mould and platen moving. During the opening mould phase, the elastic deformation of the tie bar is restored, which is the reverse process of the clamping phase. After the mould is opened, the cylinder retracts to make the platen return to its initial position, completing the platen moving phase. The two processes and four phases of the mechanism motion can be divided into the mechanism motion process and the component deformation process according to the working principle. The mould merging and separation is the moving phase of the platen, and the mould clamping and opening is the deformation phase of the component, as shown in [Table 1](#).

2.2. Kinematic modelling based on homogeneous transformation method

Kinematic models can establish the relationship between input and output of the mechanism in terms of position and velocity, and forming time-varying parameters related to component poses, which is the premise of dynamic modelling.

The MCM is a symmetrical mechanism with a double hyperelliptic-elbow, five-point, and oblique arrangement. It is simplified, and the

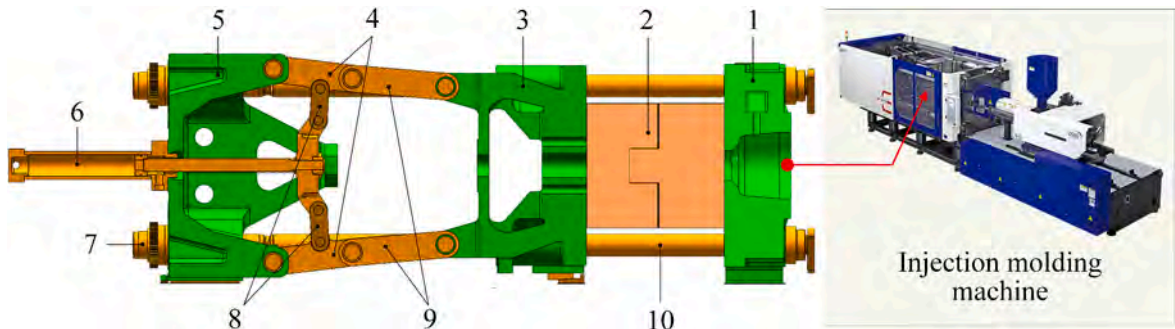


Fig. 1. The MCM of IMM (1. Front fixing platen; 2. Mould; 3. Moving platen; 4. Retrah connecting rod; 5. Retrah fixing platen; 6. Clamping cylinder; 7. Adjustment die nut; 8. Small connecting rod; 9. Front connecting rod; 10. Tie bar.)

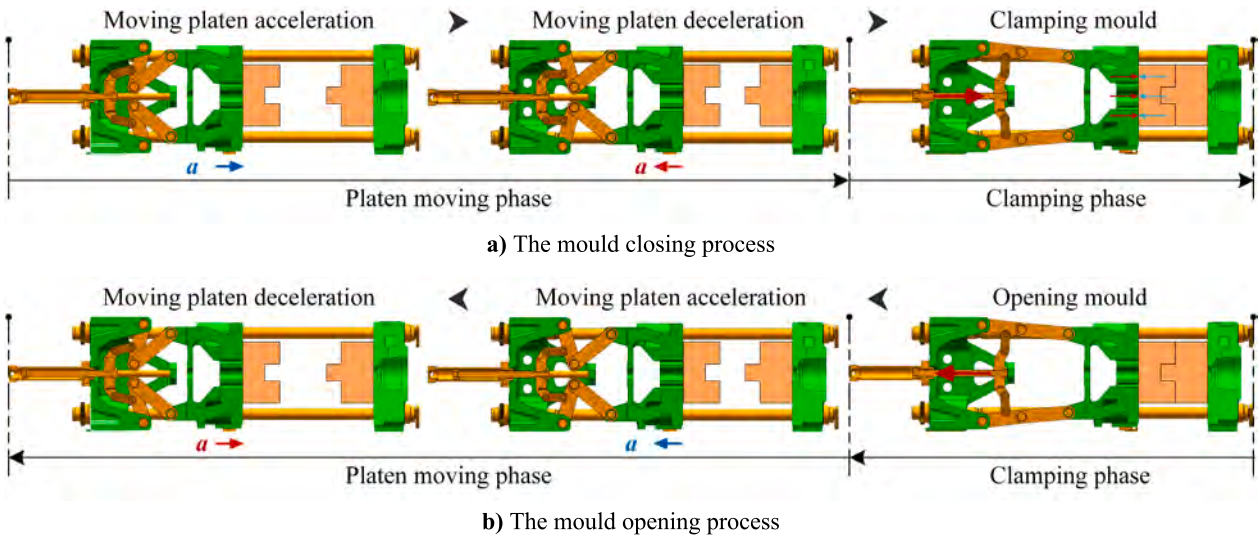


Fig. 2. Multi-phase clamping motion of the MCM.

Table 1
The working principle of each phase of the multi-phase clamping motion.

Mode	Platen moving phase	Clamping phase		Platen moving phase
	Mould merging	Mould clamping	Mould opening	Mould separation
Motion process	✓	–	–	✓
Deformation process	–	✓	✓	–

upper part of the mechanism is selected for analysis. During the motion process, both the position and orientation of the components change simultaneously. Establishing spatial coordinate systems on each joint of the mechanism allows the transformation of the motion relationships of the mechanism into numerical relationships in spatial coordinate systems, forming a comprehensive expression of pose changes and their interactions. The mechanism diagram and joint reference coordinate systems are shown in Fig. 3.

The MCM pivots around the fixed joint point O. Input force is provided through the hydraulic cylinder (simplified as slider A), transmitted through the component OBC rotation, driving the platen (simplified as slider D) in motion. Therefore, the mechanism is divided into two groups of crank-slider mechanisms, OBA and OCD, and the kinematic model is established by jointly solving forward and inverse kinematics. The purple and orange coordinate systems, coinciding with the origin O of the base coordinate system (in black), are the joint co-

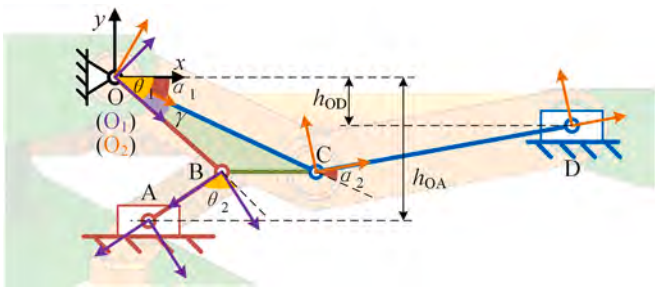


Fig. 3. The joint reference coordinate systems of the MCM.

ordinate systems of the connecting rod OB and OC, respectively. Their origins are denoted as O_1 and O_2 , coinciding with O. During the movement, the components all move the Oxy plane of the base coordinate system in Fig. 3. The transformation relationship can be expressed by the transformation matrix. The transformation matrix of rotation around the z -axis and two-dimensional plane movement can be expressed as:

$${}^P_QT = \begin{bmatrix} \cos\varphi_i & -\sin\varphi_i & 0 & L_x \\ \sin\varphi_i & \cos\varphi_i & 0 & L_y \\ 0 & 0 & 1 & L_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Solving the kinematic model requires four independent parameters of connecting rods of the mechanism: the rod length, the torsional angle between the rods, the offset between the rods, and the joint variables. Among these, only the joint variables are variables, while the rest of the parameters are constants. The corresponding parameters for forward and inverse kinematics are presented in Table 2 and Table 3, respectively.

(1) Inverse kinematics solution

The parameters in Table 2 are substituted into Eq. (1), the transformation matrix ${}^O_{O_1}T$, ${}^{O_1}_{O_2}T$ and ${}^B_A T$ between the coordinate systems of the mechanism OBA is obtained. By sequentially left-multiplying them, the pose transformation matrix for the input end A of the mechanism OBA relative to the origin O is obtained.

$$T_A^O = T_{O_1}^O \cdot T_{O_2}^{O_1} \cdot T_A^B$$

$$= \begin{bmatrix} \cos(\theta_1 + \theta_2) & -\sin(\theta_1 + \theta_2) & 0 & L_{OB}\cos\theta_1 + L_{AB}\cos(\theta_1 + \theta_2) \\ \sin(\theta_1 + \theta_2) & \cos(\theta_1 + \theta_2) & 0 & L_{OB}\sin\theta_1 + L_{AB}\sin(\theta_1 + \theta_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (2)$$

The elements corresponding to the positions (1, 4) and (2, 4) of matrix ${}^O_A T$, are the position expressions of the position coordinates of the input end A relative to the base coordinate system O. When the position coordinates of A are known as $(X_A, -h_{OA})$, the inverse kinematics equations are obtained as follows:

$$\begin{cases} L_{OB} \cdot \cos\theta_1 + L_{AB} \cdot \cos(\theta_1 + \theta_2) = X_A \\ L_{OB} \cdot \sin\theta_1 + L_{AB} \cdot \sin(\theta_1 + \theta_2) = -h_{OA} \end{cases} \quad (3)$$

(2) Forward kinematics solution

Utilizing Table 3, obtain the coordinate transformation matrices ${}^O_{O_2}T$, ${}^{O_2}_D T$, and ${}^C_D T$ for the mechanism OCD. By left-multiplying them, the pose transformation matrix of the output end D relative to the origin O is obtained.

$$T_D^O = T_{O_2}^O \cdot T_C^{O_2} \cdot T_D^C$$

$$= \begin{bmatrix} \cos(\alpha_1 + \alpha_2) & -\sin(\alpha_1 + \alpha_2) & 0 & L_{OC}\cos\alpha_1 + L_{CD}\cos(\alpha_1 + \alpha_2) \\ \sin(\alpha_1 + \alpha_2) & \cos(\alpha_1 + \alpha_2) & 0 & L_{OC}\sin\alpha_1 + L_{CD}\sin(\alpha_1 + \alpha_2) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

The elements corresponding to the positions (1, 4) and (2, 4) of matrix ${}^O_D T$, are the position expressions of the position coordinates of the output end D relative to the base coordinate system O. When the position coordinates of D are known as $(X_D, -h_{OD})$, the Forward kinematics equations are obtained as follows:

$$\begin{cases} L_{OC} \cdot \cos\alpha_1 + L_{CD} \cdot \cos(\alpha_1 + \alpha_2) = X_D \\ L_{OC} \cdot \sin\alpha_1 + L_{CD} \cdot \sin(\alpha_1 + \alpha_2) = -h_{OD} \end{cases} \quad (5)$$

After establishing the forward and inverse kinematics equations based on the base coordinate system O, the relationship between the

Table 2
Inverse kinematics solution parameters of the OBA connecting rod of the MCM.

i	Torsional angle α_i (°)	Rod length a_{i-1} (mm)	Offset distance d_i (mm)	Joint variable θ_i (°)
1	0	0	0	θ_1
2	0	$L_{OB} = 217.4$	0	θ_2
3	0	$L_{AB} = 135.5$	0	0

Table 3

Forward kinematics solution parameters of the OCD connecting rod of the MCM.

i	Torsional angle α_i (°)	Rod length a_{i-1} (mm)	Offset distance d_i (mm)	Joint variable θ_i (°)
1	0	0	0	α_1
2	0	$L_{OC} = 340.0$	0	α_2
3	0	$L_{CD} = 400.0$	0	0

time-varying parameter θ_1 and α_1 is established by the angle γ of the component OBC. The displacement mapping model of the platen about the cylinder is completed. That is:

$$\begin{cases} L_{OB} \cdot \cos\theta_1 + L_{AB} \cdot \cos(\theta_1 + \theta_2) = X_A \\ L_{OB} \cdot \sin\theta_1 + L_{AB} \cdot \sin(\theta_1 + \theta_2) = -h_{OA} \\ L_{OC} \cdot \cos\alpha_1 + L_{CD} \cdot \cos(\alpha_1 + \alpha_2) = X_D \\ L_{OC} \cdot \sin\alpha_1 + L_{CD} \cdot \sin(\alpha_1 + \alpha_2) = -h_{OD} \\ \alpha_1 = \theta_1 - \gamma \end{cases} \quad (6)$$

In Eq. (6), θ_1 , θ_2 , α_1 and α_2 are time-varying characteristic parameters describing the pose of the component. Their first and second derivatives can serve as reference variables for describing the pose changes in the dynamic model.

3. Propose dynamic modelling methods

This section is divided into three subsections to introduce the methods proposed for constructing the dynamic models of the MCM at various phases, ultimately achieving the overall process model of multi-phase motions: dynamic modelling of platen moving phase by SRM based on Newton-Euler dynamics, dynamic modelling of clamping phase by DDM based on the ratio of component stiffness coefficients, and design of phase switching logic and identification criteria based on dual parameters.

3.1. Dynamic modelling of platen moving phase by SRM based on Newton-Euler dynamics

The Platen moving phase is a pure motion phase. The platen and rod components, weighing several tons, need to achieve millimetre-level positioning accuracy within a single stroke of approximately 1 m, requiring 1 s for completion. This makes the influence of the equivalent inertia of each component and the coupling inertia between them clearly apparent. In order to analyse the coupling relationships between state variables, the model needs to include external forces, inertial forces, friction, and other factors to achieve detailed modelling. To meet the requirements of the model, this paper innovatively applies the Newton-Euler dynamics for dynamic modelling.

The Newton-Euler dynamics is commonly used in modelling scenarios such as multi-joint manipulator modelling [31,32], dynamic coupling [33], trajectory optimization, and dynamic compensation [34,35]. The drive space of a multi-joint manipulator is all located in the base coordinate system, making it a fixed space. It allows unidirectional derivation from the fixed drive space to the dynamic operating space. The difference is that the fixed space of the mechanism in this paper is at the origin O of the joint space on the power transmission route, while the driving space is in the dynamic space. The numerical chain of the equation is interrupted by the fixed joint space, leading to the inability to achieve a unidirectional continuous derivation. Therefore, this paper proposes the SRM for dynamic modelling based on Newton-Euler dynamics, as illustrated in the process shown in Fig. 4.

The SRM starts from the joint space origin point O and symmetrically extends outward to the driving space A and the operating space D. It obtains the combined forces (torques) of each joint component, then introduces external forces and transfers them back to the joint space O. The supporting force in the driving space and the required force in the

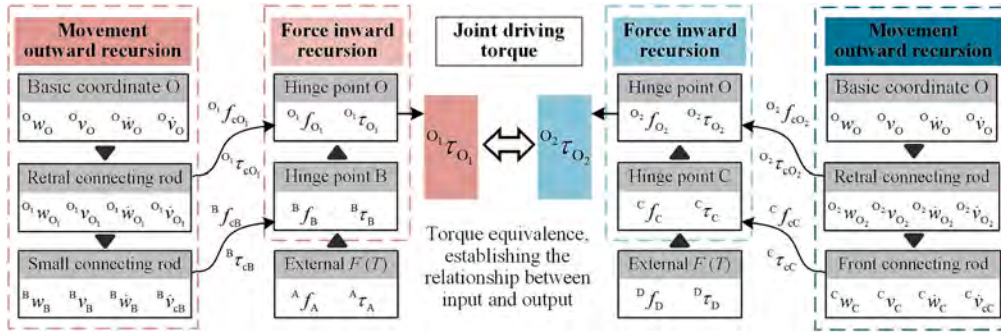


Fig. 4. The SRM solving process based on Newton-Euler dynamics.

operating space are related in the joint space base coordinates O . Finally, the operation space output force, expressed in terms of input forces in the drive space and time-varying characteristic parameters in the joint space, is obtained. This establishes a dynamic model, resolving the issue of the inability to unidirectionally derive solutions in a fixed joint space. This method requires component constant information, which is obtained by quality assessment of the mechanism model through 3-D software, as shown in Table 4.

In Fig. 3, the angular velocity of the fixed joint space O is ${}^0w_O = [0 \ 0 \ 0]^T$, the angular acceleration is ${}^0\dot{w}_O = [0 \ 0 \ 0]^T$, and the linear velocity is ${}^0v_O = [0 \ 0 \ 0]^T$. Considering the gravity of the component, the linear acceleration is ${}^0\dot{v}_O = [0 \ g \ 0]^T$. The mechanism OBA decomposed from Fig. 3 is shown in Fig. 5. Joints O , B and A are all revolute joints. The equations for outward extrapolation of angular velocity, angular acceleration, and linear acceleration for revolute joints are as follows:

$$\begin{cases} {}^{i+1}w_{i+1} = {}^{i+1}R \cdot {}^i w_i + \dot{\theta}_{i+1} \cdot {}^{i+1}z_{i+1} \cdot {}^{i+1}w_{i+1} \\ {}^{i+1}\dot{w}_{i+1} = {}^{i+1}R \cdot {}^i \dot{w}_i + {}^{i+1}R \cdot {}^i w_i \times \dot{\theta}_{i+1} \cdot {}^{i+1}z_{i+1} + \ddot{\theta}_{i+1} \cdot {}^{i+1}z_{i+1} \cdot {}^{i+1}w_{i+1} \\ {}^{i+1}\dot{v}_{i+1} = {}^{i+1}R \cdot \left[{}^i \dot{v}_i + {}^i \dot{w}_i \times {}^i p_{i+1} + {}^i w_i \times ({}^i w_i \times {}^i p_{i+1}) \right] \end{cases} \quad (7)$$

where ${}^i w_i$, ${}^i \dot{w}_i$, and ${}^i \dot{v}_i$ represent the angular velocity, angular acceleration, and linear acceleration of rod i in the coordinate system $\{i\}$. ${}^{i+1}R$ is

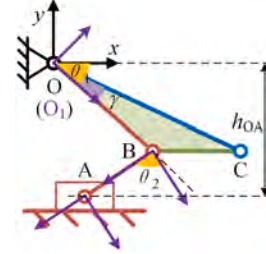


Fig. 5. The OBA mechanism joint and its coordinate system of the MCM.

rotation matrix. ${}^i p_{i+1}$ is the position vector of the origin of the coordinate system $\{i+1\}$ relative to $\{i\}$. The joint rotation angles θ_i , $\dot{\theta}_i$ and $\ddot{\theta}_i$ are set as the joint angular velocity and angular acceleration, respectively.

For the centroid of the connecting rod, the centroid acceleration is:

$${}^{i+1}\dot{v}_{c(i+1)} = {}^{i+1}\dot{v}_{i+1} + {}^{i+1}\dot{w}_{i+1} \times {}^{i+1}p_{c(i+1)} + {}^{i+1}w_{i+1} \times ({}^{i+1}w_{i+1} \times {}^{i+1}p_{c(i+1)}) \quad (8)$$

The force and torque on the centroid of the connecting rod is:

$$\begin{cases} {}^{i+1}f_{c(i+1)} = m_{i+1} \cdot {}^{i+1}\dot{v}_{c(i+1)} + {}^{i+1}w_{i+1} \times (m_{i+1} \cdot {}^{i+1}v_{c(i+1)}) \\ {}^{i+1}\tau_{c(i+1)} = {}^{ci+1}I_{i+1} \cdot {}^{i+1}\dot{w}_{i+1} + {}^{i+1}w_{i+1} \times ({}^{ci+1}I_{i+1} \cdot {}^{i+1}w_{i+1}) \end{cases} \quad (9)$$

where ${}^{i+1}f_{c(i+1)}$ and ${}^{i+1}\tau_{c(i+1)}$ are the resultant external force vector and torque vector on the connecting rod $\{i+1\}$, respectively. m_{i+1} is the mass of the connecting rod $\{i+1\}$. ${}^{ci+1}I_{i+1}$ is the inertia tensor of the connecting rod $\{i+1\}$ in the coordinate system $\{C\}$ with respect to the centroid ${}^{i+1}v_{c(i+1)}$ is the centroid linear velocity of the connecting rod $\{i+1\}$ in the coordinate system $\{i+1\}$.

From Eqs. (7), (8) and (9), the resultant external force vector on the connecting rod AB is obtained as follows:

Table 4
Component attribute information of the MCM.

Name	Material	Density	Weight	Centroid coordinate	Inertia tensor
Retral connecting rod	Nodular cast-iron (QT500)	7250 kg/m ³	$m_1 = 72.20$ kg	${}^{01}P_{c01} = \begin{bmatrix} 0.115 \\ -0.009 \\ 0 \end{bmatrix}$	${}^{c0}I_0 = \begin{bmatrix} 1.736 & -0.035 & 0 \\ -0.035 & 2.524 & 0 \\ 0 & 0 & 0.921 \end{bmatrix}$
Small connecting rod			$m_2 = 11.94$ kg	${}^{B}P_{cB} = \begin{bmatrix} 0.068 \\ 0 \\ 0 \end{bmatrix}$	${}^{cB}I_B = \begin{bmatrix} 0.019 & 0 & 0 \\ 0 & 0.024 & 0 \\ 0 & 0 & 0.006 \end{bmatrix}$
Front connecting rod			$m_3 = 70.71$ kg	${}^{C}P_{cC} = \begin{bmatrix} 0.097 \\ 0 \\ 0 \end{bmatrix}$	${}^{cC}I_C = \begin{bmatrix} 0.532 & 0 & 0 \\ 0 & 1.236 & 0 \\ 0 & 0 & 0.405 \end{bmatrix}$
Moving platen			$m_D = 1160.00$ kg	–	–

where, α_i , $\dot{\alpha}_i$, and $\ddot{\alpha}_i$ represent the joint angles, angular velocities, and

$$\begin{aligned} {}^B f_{cBx} &= m_2 g \cdot \sin(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\sin\theta_2 \cdot \ddot{\theta}_1 - 2\cos\theta_2 \cdot \dot{\theta}_1^2 - \cos\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) - m_2 \cdot {}^B p_{cBy} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^B p_{cBx} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ {}^B f_{cBy} &= m_2 g \cdot \cos(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 + 2\sin\theta_2 \cdot \dot{\theta}_1^2 + \sin\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) + m_2 \cdot {}^B p_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^B p_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \end{aligned}$$

$${}^B f_{cB} = \begin{bmatrix} {}^B f_{cBx} & {}^B f_{cBy} & 0 \end{bmatrix}^T \quad (10)$$

where ,

The resultant external torque vector on the connecting rod AB is:

$${}^B \tau_{cB} = \begin{bmatrix} 0 & 0 & {}^B I_{zz} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) \end{bmatrix}^T \quad (11)$$

The joint point A of the small connecting rod is driven by the external force and torque of the cylinder along the x -axis, and the external force and torque are set to be:

$$\begin{cases} f_A = [F_{Ax} & 0 & 0]^T \\ \tau_A = [0 & 0 & 0]^T \end{cases} \quad (12)$$

Propagating the contact forces and torques between the rods from driving space A to joint space O, there is:

$$\begin{aligned} {}^i f_i &= {}^i_{i+1} R^{i+1} f_{i+1} + {}^i f_{ci} \tau_i \\ &= {}^i_{i+1} R^{i+1} \tau_{i+1} + {}^i \tau_{ci} + {}^i p_{ci} \times {}^i f_{ci} + {}^i p_{i+1} \times {}^i_{i+1} R^{i+1} f_{i+1} \end{aligned} \quad (13)$$

According to Eq. (13), the supporting torque of the driving force provided by the driving space at the joint O of the joint space is:

$${}^O_1 \tau_{O_1} = {}^O_1 R \cdot {}^B \tau_B + {}^O_1 \tau_{cO_1} + {}^O_1 p_{cO_1} \times {}^O_1 f_{cO_1} + {}^O_1 p_B \times {}^O_1 R \cdot {}^B f_B \quad (14)$$

The mechanism OCD decomposed from the joint space O to the operation space D is shown in Fig. 6. Assuming the load of the platen as the external force in the operation space, the calculation process is the same as the principle of the OBA, and will not be elaborated here. Under the motion of the load force F_{Dx} of the platen in the operation space, the required torque reflected by the joint space joint O is calculated as follows:

$${}^O_2 \tau_{O_2} = {}^O_2 R \cdot {}^C \tau_C + {}^O_2 \tau_{cO_2} + {}^O_2 p_{cO_2} \times {}^O_2 f_{cO_2} + {}^O_2 p_C \times {}^O_2 R \cdot {}^C f_C \quad (15)$$

By the above derivation, the dynamic model of the platen moving phase in the MCM represented by the time-varying parameters of the components, is finally obtained as follows (the detailed derivation process and expression of the decomposition mechanism OBA and OCD of the elbow mechanism are shown in Appendix A):

$$F_{Dx} = f(\theta_i, \dot{\theta}_i, \ddot{\theta}_i, \alpha_i, \dot{\alpha}_i, \ddot{\alpha}_i) \quad (16)$$

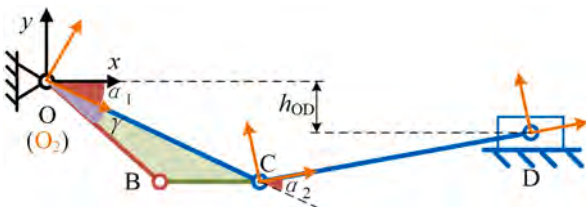


Fig. 6. The OCD mechanism joint and its coordinate system of the MCM.

angular accelerations of the mechanism OCD, respectively.

In the platen moving phase, the influence of friction on the motion is a factor that cannot be ignored. There are multi-point rotational friction and moving friction in the mechanism. The periodic oil lubrication makes the lubrication conditions of each point inconsistent, and the friction mechanism is not uniform, which brings great difficulties to the establishment of the friction model [28]. Fig. 7 shows the equivalent friction method adopted in this research, where the friction of the mechanism and the cylinder is considered as the external load F_f on the cylinder. This transforms the multi-point friction problem into a hydraulic cylinder friction problem, which is easier to test, and testing and fitting are conducted under the condition of the platen constant velocity. The diameter of the rodless chamber of the clamping cylinder is 90 mm, and the rod chamber is 55 mm. The calculation equation of the equivalent friction is:

$$P_1 \cdot A_1 - P_2 \cdot A_2 - F_f = m \cdot a \quad (17)$$

where, P_1 and P_2 are the pressure of the two chambers of the cylinder; A_1 and A_2 are the area of the two side of the cylinder; F_f is the equivalent friction force; m is the mass of the piston; a is the acceleration, and the uniform motion is 0.

To make the test results more general, the equivalent friction test incorporates factors such as oil temperature and lubrication frequency to analyse their impact on motion performance and friction conditions. The pressure of the cylinder during uniform motion is monitored hourly throughout mechanism operation, resulting in the collection of 5 sets of experimental data under varied conditions. As hydraulic pressure serves as the sole power source for the mechanism, the pressure curve reflects all encountered obstacles during motion. Fig. 8 a) depicts the experimental characteristic data alongside the corresponding equivalent friction force, illustrating the segment of the experiment involving uniform motion. Under uniform motion conditions of the cylinder, the moving platen experiences variable-speed motion, and the pressure and equivalent friction curves typically display an “arch” shape characterized by initial increase followed by decrease, attributed to the force amplification and stroke amplification properties of the MCM. The equivalent friction curve exhibits 3 distinct stages: gradual increase (blue), steep rise (pink), and subsequent decrease (green). During the blue stage, the mechanism gradually transitions from contraction to expansion, resulting in increased resistance accompanied by fluctuations in pressure and friction due to jitter. In the pink stage, as the mechanism expands, the center of gravity of the retractor connecting rod rises after passing the lowest point, leading to a steep increase in friction load accompanied by fluctuations, reaching a peak. During the green stage, the mechanism undergoes force amplification ratio growth, resulting in a slowdown of the moving platen’s speed due to force and stroke amplification. Friction-induced resistance gradually diminishes until motion cessation.

Analyse the fitting results for Fig. 8 b). It is assumed that the platen movement lubricates once every 400 times is no attenuation under lubrication conditions, and every 3000 times (operating standing parameters) is attenuated. Comparing curves ① and ②, at the beginning of the hydraulic system with a lower oil temperature and under the condition of no-decay in lubrication, the peak frictional force is

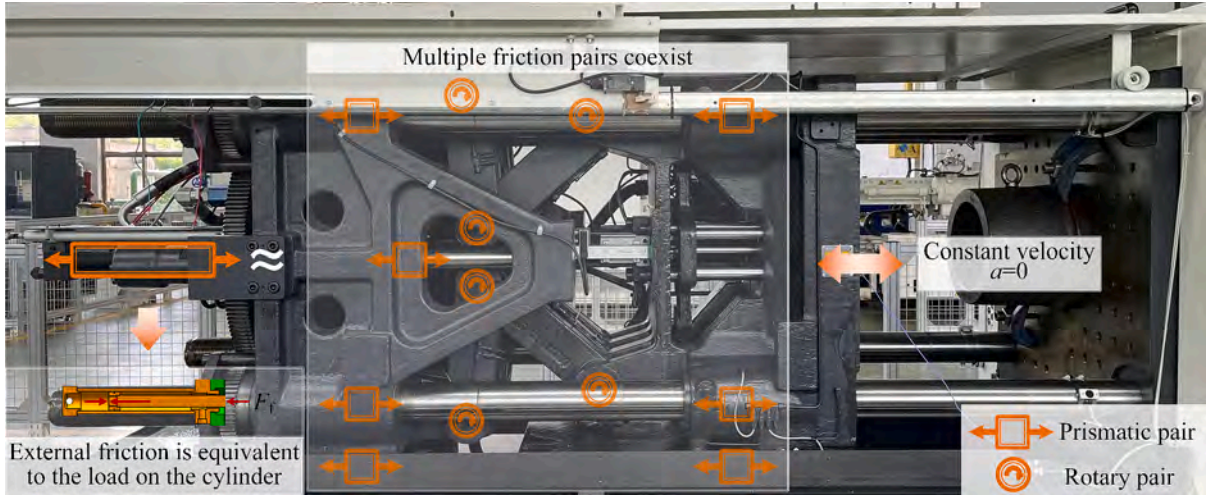


Fig. 7. Schematic diagram of the principle of the equivalent friction method.

approximately 4500 N and increases with the rise of oil temperature. When the oil temperature stabilizes at 47 °C, comparing curves ③ and ④, the equivalent friction force fluctuates in the range of 5300–6000 N under the condition of no-decay of lubrication conditions. This shows that under the condition of no-decay in lubrication, the increase in hydraulic oil temperature leads to fluctuations in equivalent frictional force. Therefore, data samples from curves ① and ② are excluded. Under the stable conditions of oil temperature (curves ③, ④, ⑤), the lubrication frequency of lubricates once every 3000 times has no obvious fluctuation compared with the equivalent friction force of the 400 times lubricate once, and the curve law is consistent. Therefore, ③, ④, ⑤ are selected as equivalent friction function fitting data samples. The fitting function and correlation evaluation results are shown in Table 5. By comparing the fitting variance R^2 , curve ⑤ exhibits the best fitting degree. This curve is selected as the equivalent friction fitting function model, and it is supplemented as a load in Eq. (16) to further improve the dynamic model.

3.2. Dynamic modelling of clamping phase by DDM based on the ratio of component stiffness coefficients

The clamping phase of the MCM is the mould merging process. When the mould is in contact, the cylinders continue to extend, pressing the platen. The deformation force generated by lengthening of the tie bar and the self-locking characteristics of the MCM are utilized to clamp the mould, completing the clamping phase. In this process, the mould undergoes compression deformation of ΔX_M under the pressure. The interacting forces of the mould cause the platen to advance only the compression distance ΔX_M under the pushing motion of the cylinder. From the kinematic model of Section 2.2, it can be inferred that the displacement value X_A of the cylinder corresponds with the displacement value X_D of the platen. When the cylinder undergoes a displacement of ΔX_A , there is a corresponding displacement of ΔX_D for the platen. Therefore, the incomplete displacement of the platen ($\Delta X_M < \Delta X_D$) will be absorbed through the elastic deformation of the components, resulting in a tensile deformation of ΔX_T in the tie bar. At the same time, the front connecting rod and the retral connecting rod undergo compression deformation of ΔX_R under the pressure. The force transmission path and elastic deformation zone during the mould clamping process are shown in Fig. 9. Based on material properties, it is assumed that the three platens are rigid components, and the clearances at joint points and small connecting rod deformations are ignored. The relationship between the theoretical displacement of the platen and the deformations of each component is as follows:

$$\Delta X_D = \Delta X_M + \Delta X_L - \Delta X_R \quad (18)$$

It can be understood that the essence of the mould clamping process is the elastic deformation of the components. The deformation of each component is allocated based on its own stiffness, where greater stiffness implies greater resistance to elastic deformation, thus determining the theoretical displacement of the platen. Therefore, this paper proposes a DDM based on the ratio of component stiffness coefficients, and establishes the clamping force model of the clamping phase. The attribute information of the main components is shown in Table 6.

During the clamping mould phase, the mould, tie bar, and connecting rod undergo varying degrees of elastic deformation due to their different stiffness. All of them can contribute to the model for representing the clamping force. In contrast, the deformation of the tie bar is uniform and easy to test. Therefore, the deformation of the tie bar is chosen as the representative measure of the clamping force. The clamping force of the MCM is characterized by four tie bars, which can be expressed as:

$$F_{Cx} = 4K_t \cdot \Delta X_D \cdot K_p \quad (19)$$

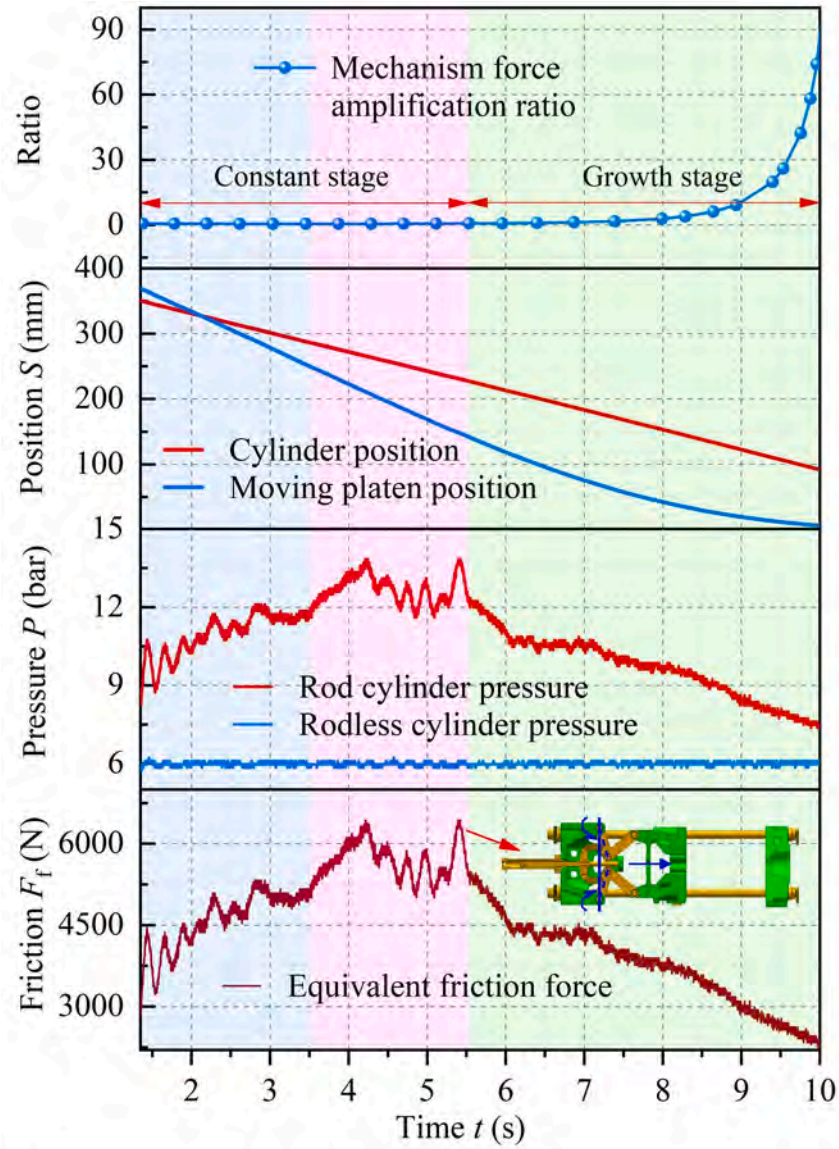
where, K_p represents the distribution ratio coefficient for the deformation of the tie bar to the theoretical displacement of the platen.

As shown in Fig. 9, the mechanism establishes a force equilibrium at the contact interface between the platen and the mould, indicating that the deformation forces of the tie bar and the connecting rod group maintain the mechanical balance with the deformation forces generated by the compression of the mould. Therefore, the three deformations can be divided into two groups by the contact surface: ΔX_M , and ΔX_T and ΔX_R . The distribution ratio coefficient K_p can be solved.

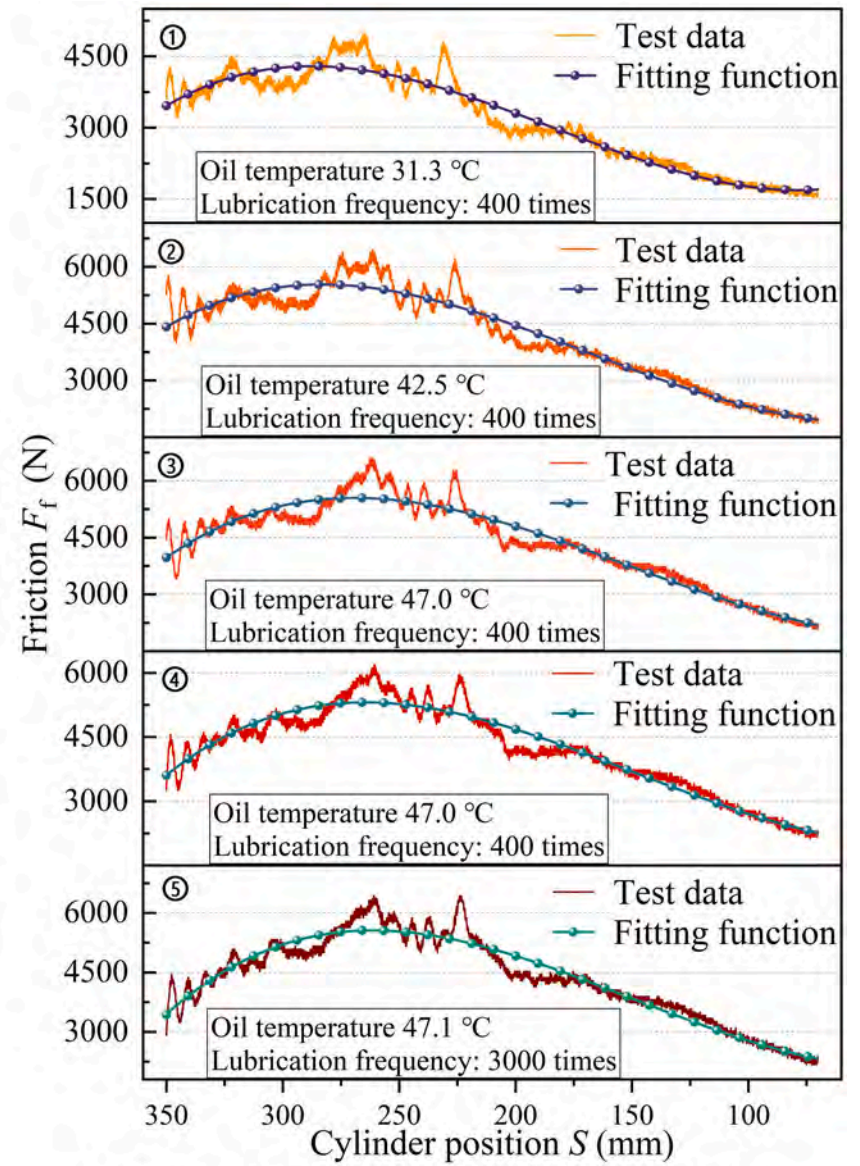
Firstly, it is assumed that K_{p1} is the distribution ratio for the distribution of the deformation of the mould, ΔX_M , relative to the theoretical displacement of the platen, ΔX_D . Using the total stiffness of the mechanism as a reference, the total stiffness is the sum of the stiffness coefficients of all deformation components. K_{p1} can be expressed as the ratio of the sum of the stiffness coefficients of all deformation components except the mould to the total stiffness. That is:

$$K_{p1} = \frac{4K_t + 2K_f + 2K_h}{4K_t + 2K_f + 2K_h + K_m} \quad (20)$$

After ΔX_M is distributed, the residual of ΔX_D is composed of the deformation of the tie bar, ΔX_T , and the deformation of the front and retral connecting rods, ΔX_R . As a representation of the clamping force, the distribution ratio for the theoretical displacement of ΔX_T is assumed to be K_{p2} . Using the total stiffness of the tie bar and the front and retral



a) Experimental characteristic data and corresponding equivalent friction



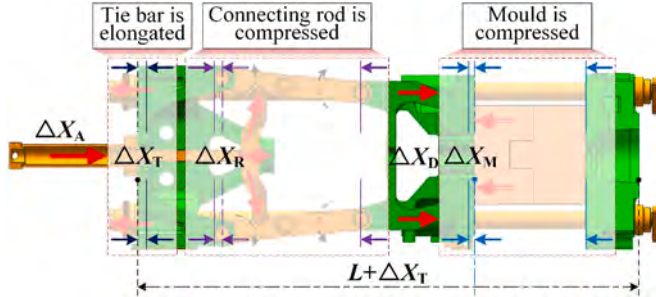
b) The fitting curves of 5 sets of equivalent friction

Fig. 8. Experimental data and fitting functions for MCM equivalent friction.

Table 5

Fitting function of equivalent friction.

Curve	Fitting function	Fitting variance
③	$F_f = 1.68 \times 10^{-8}x^3 - 4.04 \times 10^{-4}x^2 + 2.50x + 899.68$	$R^2 = 0.93334$
④	$F_f = 1.65 \times 10^{-8}x^3 - 3.84 \times 10^{-4}x^2 + 2.31x + 1154.47$	$R^2 = 0.94017$
⑤	$F_f = 1.95 \times 10^{-8}x^3 - 4.52 \times 10^{-4}x^2 + 2.77x + 467.35$	$R^2 = 0.94294$

**Fig. 9.** The mould clamping mechanism diagram of the MCM.**Table 6**

The main component attribute information table of the MCM.

Component	Material	Elastic modulus (Pa)	Section area (m ²)	Length (m)	Stiffness coefficient (kN/m)
Tie bar	42CrMoA	2.12×10^{11}	0.007	2.878	$K_t = 5.22 \times 10^5$
Mould	QT500	2.06×10^{11}	0.042	0.315	$K_m = 2.75 \times 10^7$
Front connecting rod	QT500	2.06×10^{11}	0.018	0.515	$K_f = 7.31 \times 10^6$
Retral connecting rod	QT500	2.06×10^{11}	0.023	0.455	$K_h = 1.03 \times 10^7$

connecting rods as a reference, the total stiffness is the sum of the stiffness coefficients of the tie bar and the front and retral connecting rods. K_{p2} can be expressed as the ratio of the sum of the stiffness coefficients of the front and retral connecting rods to the total stiffness coefficients of the tie bar and the front and retral connecting rods. That is:

$$K_{p2} = \frac{2K_f + 2K_h}{4K_t + 2K_f + 2K_h} \quad (21)$$

Finally, the distribution ratio coefficient K_p for the distribution of the tie bar to the theoretical displacement of the platen is determined, which is:

$$K_p = K_{p1} \cdot K_{p2} = \frac{2K_f + 2K_h}{4K_t + 2K_f + 2K_h + K_m} \quad (22)$$

After completion, considering the model error caused by assumptions and calculations, the correction coefficient ε is introduced. The error mainly comes from two parts: First, the calculation error of the stiffness coefficients of the components, as the components have non-uniform and irregular shapes. The other part is caused by the pure rigidity assumption of the other component and the ignore of the angle of the plane motion of the connecting rod. In conclusion, a comprehensive error coefficient $\varepsilon = 0.9$ is chosen to compensate for the model error. Substituting K_p and ε into Eq. (19), the clamping force model during the clamping phase is obtained as follows:

$$F_{Cx} = 4K_t \cdot \Delta X_D \cdot \frac{2K_f + 2K_h}{4K_t + 2K_f + 2K_h + K_m} \cdot \varepsilon \quad (23)$$

3.3. Design of phase switching logic and identification criteria based on dual parameters

The identification and switching during the motion phase are crucial for ensuring a smooth transition between the platen moving phase and the clamping mould phase. Currently, IMMs adopt the position single-variable identification and switching, which has low fault tolerance and accuracy. This paper proposes an identification and switching strategy based on dual parameters for recognizing transitions between motion stages. The logic and conditions are refined based on the state quantities of the MCM. The relevant state quantities and characteristics are as follows:

- Position values: The position values of the cylinder and the platen can directly reflect the position area of the mechanism and can be directly monitored. However, due to the large stiffness of the platen and mould materials, the changes in the clamping phase are not sufficiently pronounced, which can be used to evaluate the range of the switching critical point.
- Cylinder pressure value: During the clamping mould phase, the pressure value changes more significantly compared with the position value. The sensor, with its fast response speed, serves as a precise critical feature point for clamping mould, determining the exact position.
- Clamping force value: The clamping force value is the most intuitive state quantity for judging whether the motion has entered the clamping mould phase and can be directly tested. It is selected as a feature quantity to evaluate whether the recognition condition is effective.

The curves of the above state quantities in a single clamping motion are shown in Fig. 10. The figure divided the platen moving phase and the clamping mould phase through blue and pink backgrounds. Next, the characteristics of the state quantities near the critical point of clamping are analysed, and the logic and criterion are proposed.

(1) Critical point analysis of the mould merging process

The enlarged view of the critical region curve for the motion phase transitions is shown in Fig. 11, as illustrated in Fig. 10. Fig. a) represents the state curve of the critical region of the mould merging process. Before the platen approaches the critical point, it first contacts the mould, resulting in rebound. Then, it enters the low-pressure mould protection state. At this moment, the set pressure for the rodless chamber of the cylinder is 20 bar, and it is maintained with fluctuations to prevent rapid impact. During the process, the cylinder continues to advance, and the platen slowly approaches the mould. Once it contacts and fits with the mould, reaching the critical clamping point, the system switches to the clamping mould phase. During the mould clamping process, the pressure of the rodless chamber increases first and then decreases due to the force amplification characteristics of the mechanism, until the mould clamping is completed. The rod chamber is in a state of slight back pressure oil reflux. Within the mould clamping region, the position value of the platen consistently remains within a 0.5 mm range, while the pressure in the two cavities fluctuates simultaneously before the clamping process begins. After clamping, the pressure in the rodless chamber is maintained at a certain value, and there is low-pressure oil reflux in the rod chamber.

(2) Critical point analysis of the mould separation process

Fig. 11 b) is the state curve of the critical region of the mould separating process. Mould separation is the process of releasing the clamping force. Control logic switching transforms the hydraulic system into a state of large throttling and high back pressure, preventing the

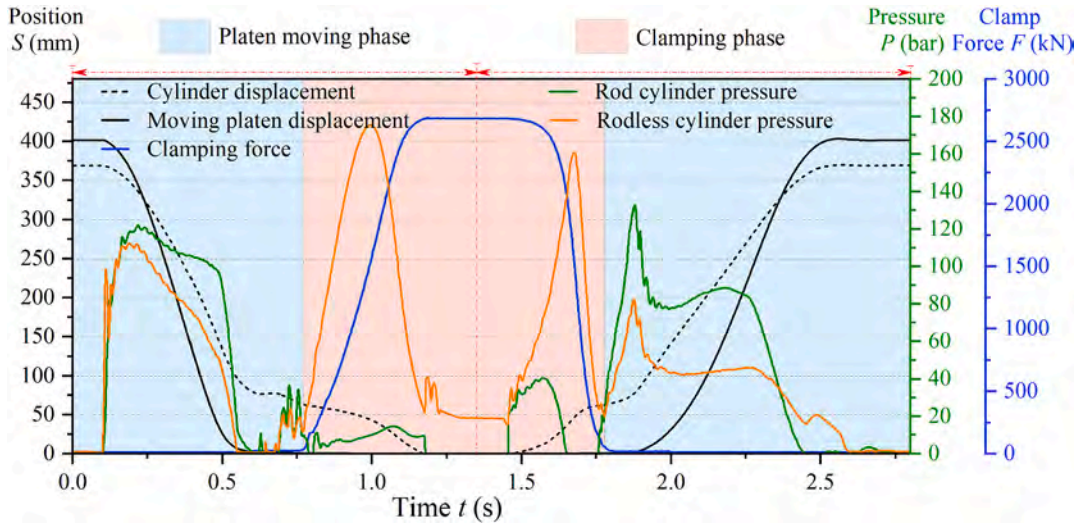


Fig. 10. State quantity curve of the MCM system.

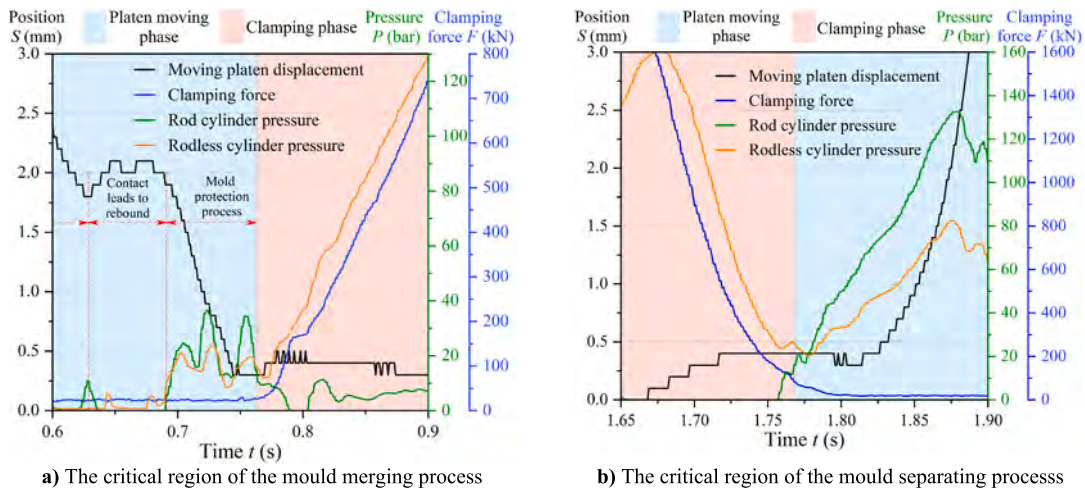


Fig. 11. State quantity curve of phase switching critical region of the MCM

elastic potential energy of the tie bar from causing the platen to go out of control. Following the mould separation, the return oil circuit of the system switches to a state of slight back pressure, and the motion is switched to the platen moving phase. In the critical region of the mould separating, the platen position value is always <0.5 mm. Throughout the process, the pressure in the rod chamber experiences an initial increase, while the pressure in the rodless chamber remains below 20 bar during the constant clamping force phase. However, the pressure in the rodless chamber immediately rises as the pressure in the rod chamber increases, continuing until the transition to the platen moving phase.

During the moving clamping phase, the position value of the platen consistently remains within a 0.5 mm range. The pressure of the rodless chamber first increases and then decreases; during the constant

clamping force phase, it remains below 20 bar. The pressure in the rod chamber, apart from the initial increase during mould separating, remains in a low-pressure state, and the fluctuations are reflected in the pressure of the rodless chamber. In summary, the switching logic is determined. The identification area is defined by the position of the platen. The rod chamber pressure value determines the switching critical point, and the special point is supplemented by the rodless chamber pressure value. The switching conditions are determined below.

According to the characteristics of the state quantity, when $X_D \leq 0.5$ mm, the platen enters the identification range of the clamping phase, and the system is in the low-pressure mould protection state. Within the range, the pressure values of the two chambers fluctuate at the same frequency. After the fluctuation stops, it enters the clamping phase, and

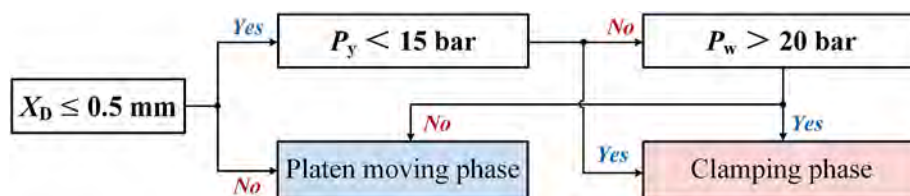


Fig. 12. The diagram of phase switching logic and identification criteria for the motion of the MCM.

the critical clamping point can be determined at the final trough point, as shown in Fig. 11 a). The pressure of the rod chamber at the trough point is affected by the fluctuation and fluctuates at 15 bar, taking 15 bar as the phase switching condition. When $P_y < 15$ bar, it is determined that the motion enters the clamping phase.

Comparing the numerical values of the critical point separation with the mould merging switching condition of $X_D \leq 0.5$ mm and $P_y < 15$ bar. The critical region of mould separation, with the platen position maintained at 0.4 mm, and the pressure value of the rod chamber begins to rise. When P_y reaches 15 bar, the decrease in clamping force value ceases, marking the identification of the critical point of mould separation. When the rod chamber is pressurized, the pressure in the rodless chamber will rise rapidly. According to this feature, the identification criterion can be augmented for specific points with $P_y > 15$ bar in the mould separation process. With a constant clamping force, the pressure in the rodless chamber is maintained at approximately 20 bar. Therefore, as a supplement to the identification criterion, when $P_y > 15$ bar, if the pressure value in the rodless chamber $P_w > 20$ bar, it is still judged that the motion is in the clamping phase. Thus, the switching logic and identification criterion for the phase switch of the MCM were designed, as shown in Fig. 12.

By combining the switching logic and identification criteria with the platen moving and the clamping phase models, a comprehensive dynamic model for the multi-stage motion of the MCM is established, namely:

$$\begin{cases} F_{Dx} = f(\theta_1, \dot{\theta}_1, \ddot{\theta}_1, \alpha_1, \dot{\alpha}_1, \ddot{\alpha}_1, F_i) & (X_D > 0.5 \text{ mm}) \\ F_{Dx} = f(\Delta X_D) \begin{cases} X_D \leq 0.5 \text{ mm and } P_y < 15 \text{ bar} \\ \text{or} \\ X_D \leq 0.5 \text{ mm, } P_y \geq 15 \text{ bar and } P_w > 20 \text{ bar} \end{cases} & (24) \end{cases}$$

4. Experimental testing and results analysis

To validate the effectiveness of the method and model, an experimental platform was constructed, and experimental tests were

conducted. Fig. 13 shows a three-plate IMM testbed, which includes an IMM, sensor, signal integration module, EtherCAT field bus, acquisition interaction platform and so on. Data from the clamping motion mechanism is collected through sensors, and the integrated signals are transmitted via the EtherCAT fieldbus system. Experimental data is acquired on the data acquisition and interaction platform. The experimental process for the MCM includes continuous automatic motion tests and manual uniform motion tests. The relevant parameters involved in the test are detailed in Table 7 (The percentage in the table represents the percentage of the highest speed).

4.1. Validation for the switching logic and identification criterion between motion phases

To validate the effectiveness of the phase switching logic and identification criterion, this paper introduces the concept of the clamping force rate. The clamping force rate measures the rate of increase in clamping force during the clamping mould phase, and the clamping force in Fig. 10 is calculated by the difference method, as shown in Fig. 14.

In the Enlarged view (1) of Fig. 14, after the critical point of mould merging, the clamping force rate increases by approximately 6 kN/ms within 20 ms. Before the critical point of mould separation in the Enlarged view (2), the clamping force rate decreases by 4 kN/ms within 20 ms. Due to the fluctuation of the test data and the absence of fixed characteristic points, this paper stipulates that the critical point identified by the conditions is when the clamping force rate is < 4 kN/ms, and the absolute value of the clamping force rate within ± 20 ms is > 4 kN/ms, indicating a successful identification. Multiple operating conditions are run on the IMM testbed based on positional values and dual parameters respectively, with 30 sets of data extracted for each. The effectiveness of the proposed switching logic and conditions is compared and verified, as shown in Figs. 15 and 16.

According to the identification results of Figs. 15 and 16, it is evident that under two recognition conditions, 60 critical points are identified respectively. Among them, with the single-parameter based on positional values, 13 did not meet the criteria, resulting in a recognition

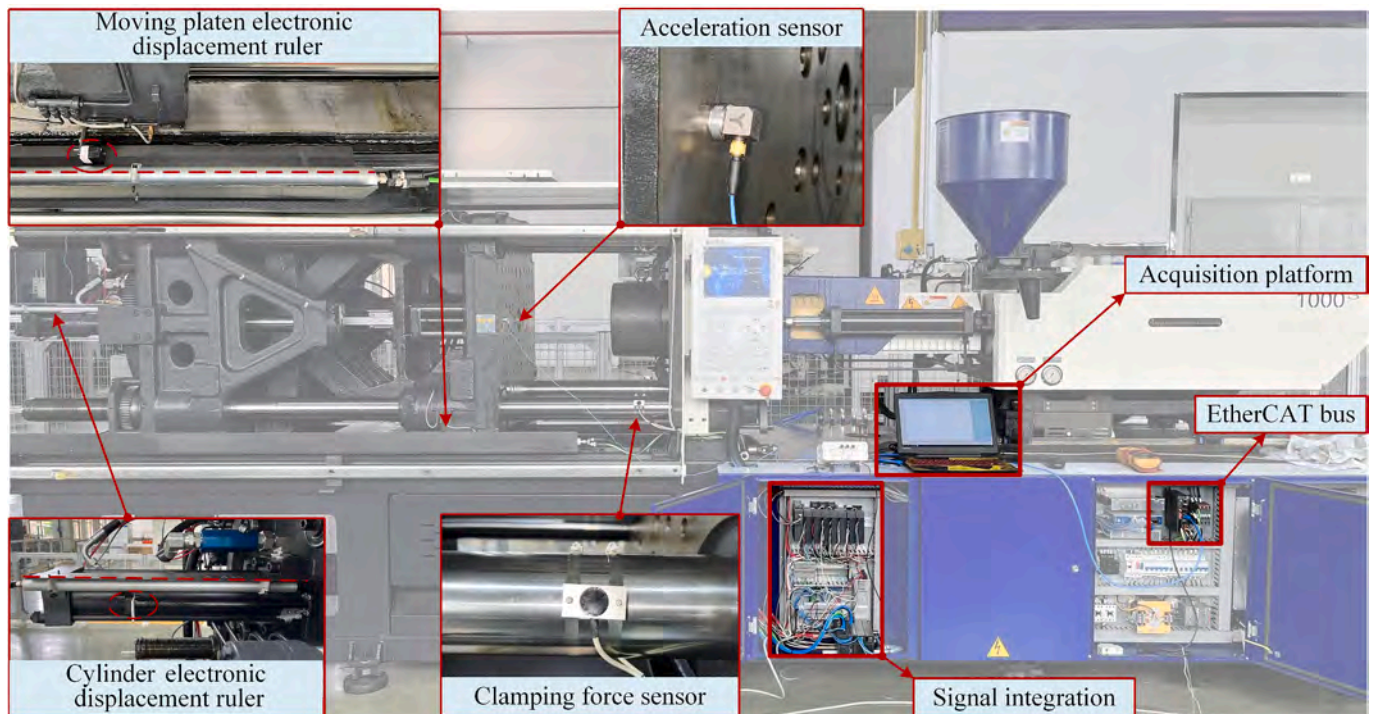


Fig. 13. The IMM testbed.

Table 7
Parameters of the MCM.

Platen moving phase	Merging process	Separating process	Clamping phase	Merging process	Separating process
Displacement (mm)	400	400	Position (mm)	5.0	3.0
Velocity	99 %	99 %	Pressure (bar)	175	175
Compensation (mm)	4.3	4.3	Velocity	45 %	45 %

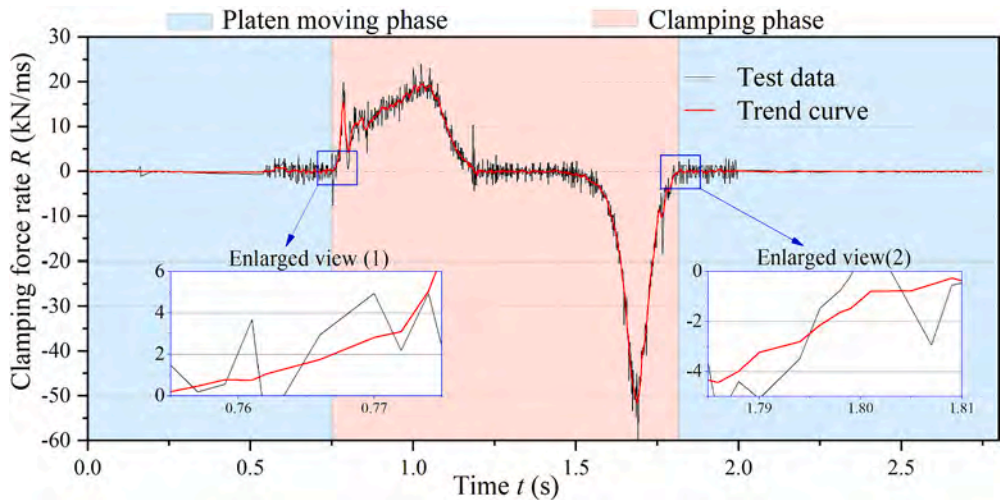


Fig. 14. Clamping force rate of the MCM.

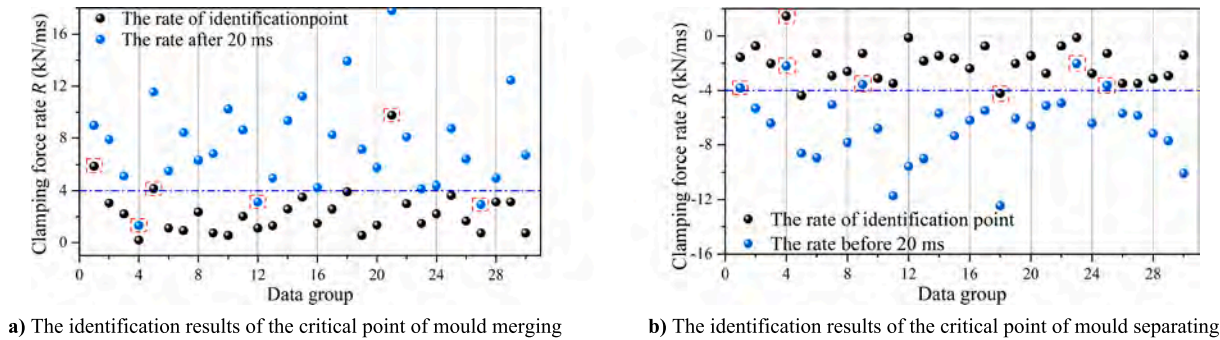


Fig. 15. The phase switching and recognition results based on single-parameter positional values.

success rate of 78.3 %. With the dual-parameter-based approach, only 3 did not meet the criteria, achieving a recognition success rate of 95 %. This demonstrates the superiority of the proposed phase switching logic and identification criteria.

4.2. Validation for the dynamic model and modelling methods

The continuous automatic motion test process is shown in Fig. 17. The test was carried out for 6 h, including various working conditions of different oil temperatures and different lubrication frequencies. To ensure the universality of the verification results, three sets of data samples containing various states were extracted from the test data for

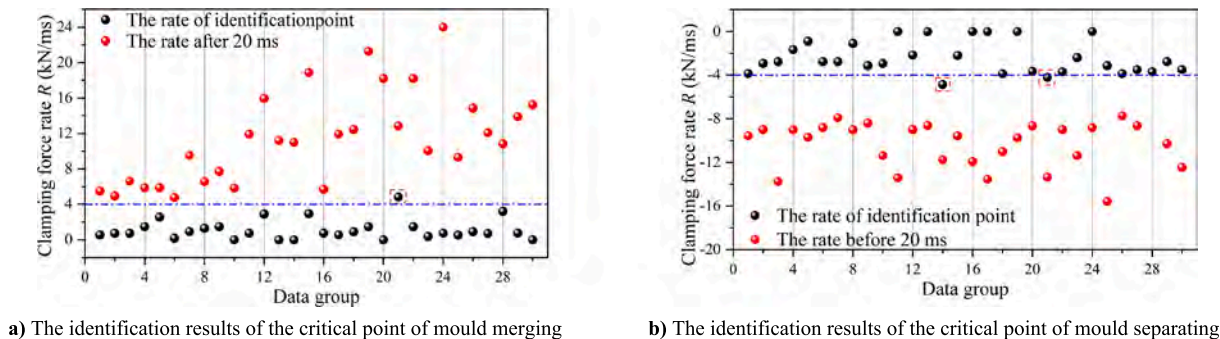


Fig. 16. The phase switching and recognition results based on positional-pressure dual parameters.

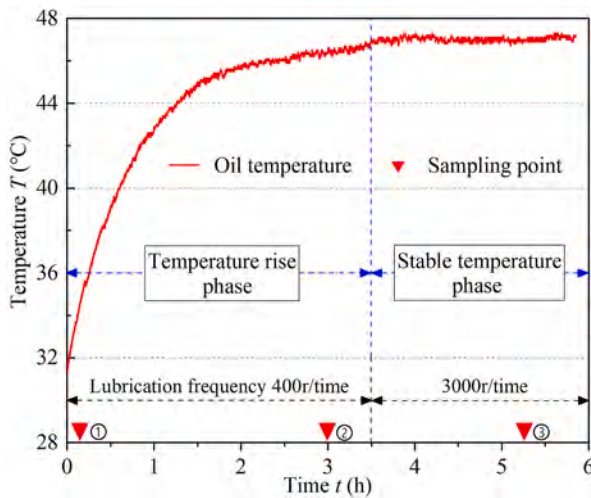


Fig. 17. Experimental testing process.

analysis. The correlation coefficient R^2 of the residual sum of squares method is employed to evaluate the error between the model quantity and the test quantity, reflecting the correlation between the curves, which can be expressed as:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (25)$$

where, n is the number of test samples. y_i and $\hat{y}_i$ represent the model value and test value of the test point, respectively. $\bar{y}$ is the mean value of the model value. Essentially, R^2 represents the correlation between the model value and the test value. The closer R^2 is to 1, the higher the correlation is. In the verification, it is assumed that when $R^2 > 0.8$, the model value and the test value have a good correlation.

Taking the displacement of the cylinder as the input boundary condition of the model, the multi-phase dynamic model Eq. (24) is run by mathematical operation software, and the correlation between the model and the test quantity is compared. The comparison of the correlation between the model quantities and test quantities for three sets of samples is shown in Figs. 19–20, and the correlation results are presented in Tables 8–10.

The sample ① is the initial stage of the testbed operation. With a

hydraulic system having low oil temperature and high lubrication frequency, which is the optimal stage for the performance of the IMM. In Fig. 18, the platen moving and clamping phases of the sample are divided into 11 phases according to the critical point. The odd number phases (1), (3) ... (11) is the platen moving phase. The model considers the inertia of the component and the coupling inertia between the components, and achieves a good description of the dynamic trends of the platen. The correlation between the model curve and the test results is 92–95 %. The even number phases (2), (4) ... (10) are the clamping phases. The construction mechanism and results of the clamping force are accurately described by the deformation of the tie bar, and the correlation is above 97 %. Finally, the correlation evaluation of the entire curve was carried out. Sample ① achieved 99.19 %, and the dynamic trend was very consistent.

The sample ② is taken after the test bench has been running for a period, during which the system oil temperature increases, causing a decline in control performance and a decrease in motion accuracy. Based on the above situation, the correlation between the model values and test results in the odd-numbered phases of Fig. 19 is 90–95 %, and the fluctuation range becomes larger. Compared with the test results, the model values of the even-number phases have errors at some peak values of the clamping force, but still accurately describe the change process of the clamping force, and the correlation remains above 97 %. The correlation of the entire curve reaches 98.42 %, which is slightly lower compared to the initial operational phase, but the coincidence is very good.

The sample ③ is that after the oil temperature of the testbed is stabilizes, the lubrication is adjusted to the normal frequency, and the performance tends to be stable and no longer declines. In this state, the correlation between the model values of odd-numbered phases in Fig. 20 and test results remains above 90 %. The correlation of the even-number phases remained above 97 %. The correlation between the entire curve and the test results is 97.63 %, which is slightly lower compared to the previous two data samples.

The above results analysis indicates that based on the modelling method and phase recognition and switching logic proposed in this study, the model relevance can be maintained at above 97 % under various operating conditions. The MCM previously employed a model based on the principle of virtual displacement, as shown in Eq. (26). This method can only describe and control the clamping phase based on position information and force amplification ratio (even-numbered stages in Figs. 18–20), while the moving platen phase is open-loop controlled, relying solely on the kinematic model based on uniform

Table 8

The correlation coefficient between the model and the test quantity of sample ①.

Sub-phase	(1)	(3)	(5)	(7)	(9)	(11)	The whole curve
R^2	94.78 %	93.7 %	93.65 %	93.2 %	92.7 %	92.43 %	99.19 %
Sub-Phase	(2)	(4)	(6)	(8)	(10)	–	
R^2	98.76 %	97.89 %	98.48 %	99.10 %	99.47 %	–	

Table 9

The correlation coefficient between the model and the test quantity of sample ②.

Sub-phase	(1)	(3)	(5)	(7)	(9)	(11)	The whole curve
R^2	94.99 %	95.03 %	95.84 %	92.57 %	93.58 %	91.14 %	98.42 %
Sub-Phase	(2)	(4)	(6)	(8)	(10)	–	
R^2	97.23 %	98.12 %	98.19 %	99.31 %	99.35 %	–	

Table 10

The correlation coefficient between the model and the test quantity of sample ③.

Sub-phase	(1)	(3)	(5)	(7)	(9)	(11)	The whole curve
R^2	94.89 %	93.23 %	92.49 %	92.08 %	93.00 %	90.85 %	97.63 %
Sub-Phase	(2)	(4)	(6)	(8)	(10)	–	
R^2	97.22 %	97.47 %	99.34 %	97.84 %	98.97 %	–	

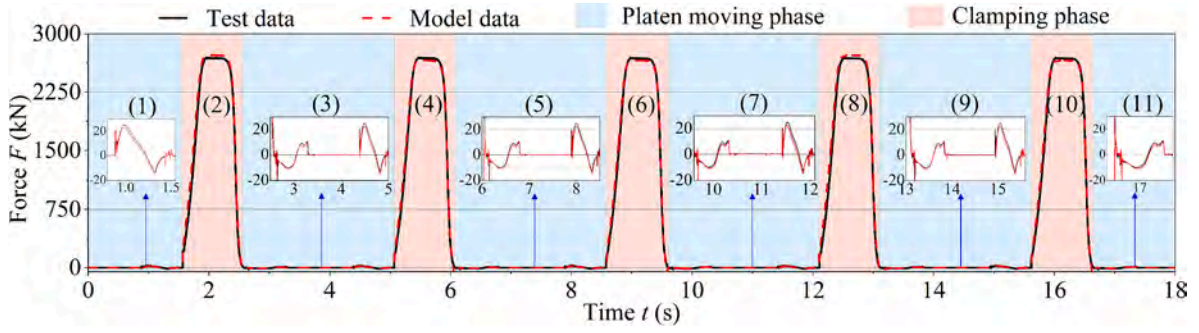


Fig. 18. Comparison of correlation between model quantity and test quantity of sample ①.

variable motion ($X = V_{ot} + \frac{1}{2}at^2$), incapable of completing the dynamic description of the platen. Compared with the previous method, this study not only achieves the dynamic description of the entire process but also demonstrates a higher correlation ratio, reflecting the effectiveness and reliability of the proposed method and model.

$$F_{Dx} = \frac{\Delta X_{A(i+1)} - \Delta X_{A(i)}}{\Delta X_{D(i+1)} - \Delta X_{D(i)}} (P_1 \cdot A_1 - P_2 \cdot A_2) \quad (26)$$

5. Conclusion and discussion

This research focuses on the multi-phase MCM of IMMs and the distinct characteristics of their motion mechanisms, alongside the nonlinear effects of inertia, friction, and deformation when constructing models. We studied the dynamic model of the entire process and validated the effectiveness and reliability of our approach. The specific innovative contributions are outlined as follows:

(1) In response to the influence of inertia and friction during the platen moving phase of the MCM, as well as the problem of low motion accuracy caused by the kinematic model, a SRM based on Newton-Euler dynamics is proposed. By designing equivalent friction experiments, the

impact of multi-point friction is addressed, enabling the construction of a high-precision model. This presents the dynamic state of the platen moving phase and the SRM resolves the issue of fixed joint space mechanisms being unable to unilaterally deduce solutions.

(2) Addressing the nonlinear effects of deformation of components during the clamping phase, we propose a DDM based on the ratio of component stiffness coefficients. Leveraging the principle that higher component stiffness corresponds to more challenging elastic deformation, we establish a mapping relationship between the position of the cylinder and the deformation of components. This addresses the current limitation of estimating the clamping force using the force amplification ratio during the clamping phase, enabling accurate calculation of the clamping force value.

(3) To tackle the low accuracy of single-parameter recognition and switching based on position values during multi-phase motion recognition and switching, we design logic based on dual parameters of position and pressure. This results in a 16.7 % improvement in accuracy. Our recognition logic connects the dynamic models of the platen moving and clamping phases, forming an entire process dynamic model of the MCM.

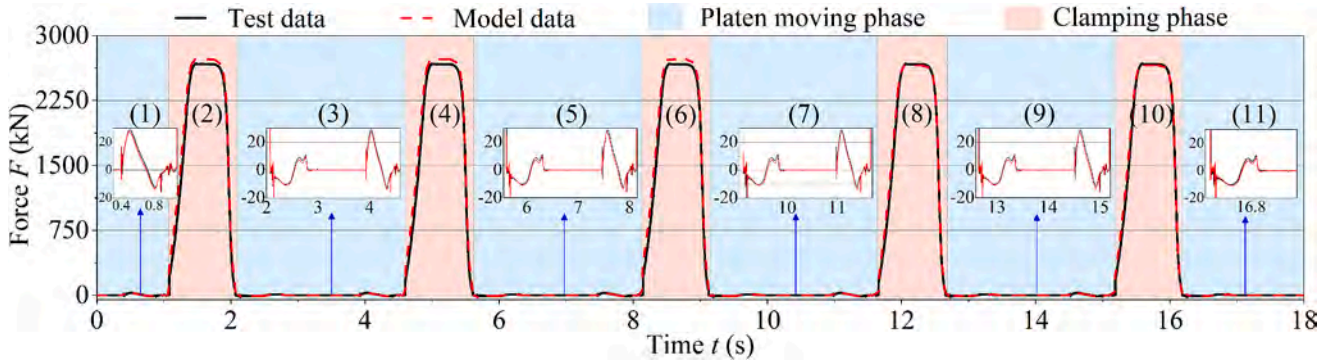


Fig. 19. Comparison of correlation between model quantity and test quantity of sample ②.

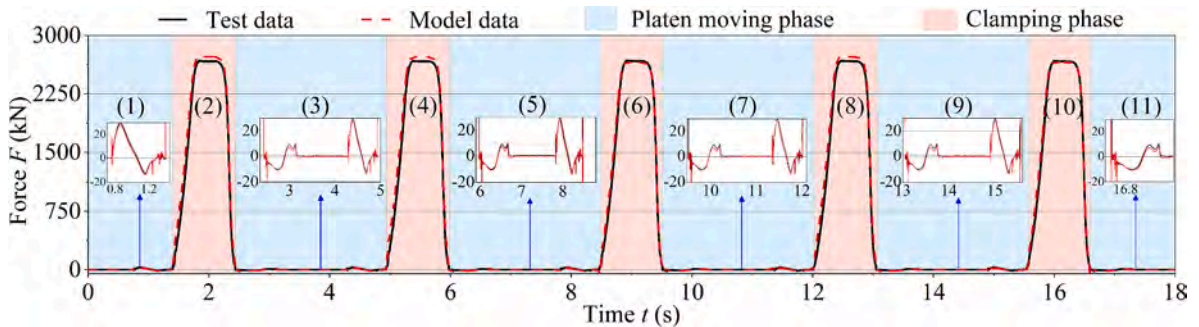


Fig. 20. Comparison of correlation between model quantity and test quantity of sample ③.

We validate the correlation between the dynamic model and experimental results, with each phase model achieving correlations of over 90 %, and the entire process model maintaining correlations above 97 % in multiple tests. This demonstrates the effectiveness and superiority of the proposed methodology and model. In the future, we will use this model and method to optimize the control of MCM, so as to improve the control performance of IMMs.

CRediT authorship contribution statement

Zhiqiang He: Conceptualization, Data curation, Methodology, Software, Writing – original draft, Writing – review & editing. **Fugang Zhai:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Writing – original draft. **Yin Zhang:** Data curation, Software, Validation, Visualization. **Weibo Wang:** Data curation, Software, Validation. **Dong Li:** Software, Validation, Writing – original draft. **Ningdi Zhu:** Conceptualization, Investigation, Validation. **Jianbo Ruan:**

Funding acquisition, Resources, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Taking the mechanism OBA as an example, the detailed derivation process of dynamics is as follows

According to Eq. (1), the rotation matrix from coordinate system $\{O_1\}$ to $\{O\}$, $\{B\}$ to $\{O_1\}$, $\{A\}$ to $\{B\}$ can be obtained. That are:

$${}^O_{O_1}R = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^O_B R = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 \\ \sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^B_A R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (A1)$$

Based on the orthogonal property of rotation matrix, each matrix of Eq. (A1) is transposed. The rotation matrix of $\{O\}$ to $\{O_1\}$, $\{O_1\}$ to $\{B\}$, $\{B\}$ to $\{A\}$ are as follows:

$${}^{O_1}_O R = \begin{bmatrix} \cos\theta_1 & \sin\theta_1 & 0 \\ -\sin\theta_1 & \cos\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^{B}_{O_1} R = \begin{bmatrix} \cos\theta_2 & \sin\theta_2 & 0 \\ -\sin\theta_2 & \cos\theta_2 & 0 \\ 0 & 0 & 1 \end{bmatrix} {}^A_B R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (A2)$$

From Fig. 5, the position vector between $\{O\}$ and $\{O_1\}$ is ${}^O p_{O_1} = [0 \ 0 \ 0]^T$, the position vector between $\{O_1\}$ and $\{B\}$ is ${}^{O_1} p_B = [L_{OB} \ 0 \ 0]^T$, the position vector between $\{B\}$ and $\{A\}$ is ${}^B p_A = [L_{BA} \ 0 \ 0]^T$.

(1) The dynamics are derived outward

Given the above conditions, the angular velocity of the retral connecting rod in the coordinate system $\{O_1\}$ is:

$${}^{O_1} w_{O_1} = [0 \ 0 \ \dot{\theta}_1]^T \quad (A3)$$

The angular acceleration of the retral connecting rod in the coordinate system $\{O_1\}$ is:

$${}^{O_1} \dot{w}_{O_1} = [0 \ 0 \ \ddot{\theta}_1]^T \quad (A4)$$

The linear velocity of the retral connecting rod in the coordinate system $\{O_1\}$ is:

$${}^{O_1} v_{O_1} = [0 \ 0 \ 0]^T \quad (A5)$$

The linear acceleration of the retral connecting rod in the coordinate system $\{O_1\}$ is:

$${}^{O_1} \dot{v}_{O_1} = [g \cdot \sin\theta_1 \ g \cdot \cos\theta_1 \ 0]^T \quad (A6)$$

The centroid linear acceleration of the retral connecting rod in the coordinate system $\{O_1\}$ is:

$${}^{O_1} \dot{v}_{cO_1} = \begin{bmatrix} g \cdot \sin\theta_1 - {}^{O_1} p_{cO_1y} \cdot \ddot{\theta}_1 - {}^{O_1} p_{cO_1x} \cdot \dot{\theta}_1^2 \\ g \cdot \cos\theta_1 + {}^{O_1} p_{cO_1x} \cdot \ddot{\theta}_1 - {}^{O_1} p_{cO_1y} \cdot \dot{\theta}_1^2 \\ 0 \end{bmatrix} \quad (A7)$$

Then, the resultant force vector on the retral connecting rod is:

$${}^I f_{c1} = \begin{bmatrix} m_1 g \cdot \sin\theta_1 - m_1 {}^{O_1} p_{cO_1y} \cdot \ddot{\theta}_1 - m_1 {}^{O_1} p_{cO_1x} \cdot \dot{\theta}_1^2 \\ m_1 g \cdot \cos\theta_1 - m_1 {}^{O_1} p_{cO_1x} \cdot \ddot{\theta}_1 - m_1 {}^{O_1} p_{cO_1y} \cdot \dot{\theta}_1^2 \\ 0 \end{bmatrix} \quad (A8)$$

The resultant torque vector on the retral connecting rod is:

$${}^{O_1}\tau_{cO_1} = \begin{bmatrix} 0 & 0 & {}^{O_1}I_{ZZ}\ddot{\theta}_1 \end{bmatrix}^T \quad (A9)$$

For the small connecting rod, the angular velocity in the coordinate system {B} is:

$${}^Bw_B = \begin{bmatrix} 0 & 0 & \dot{\theta}_1 + \dot{\theta}_2 \end{bmatrix}^T \quad (A10)$$

The angular acceleration of the small connecting rod in the coordinate system {B} is:

$${}^B\dot{w}_B = \begin{bmatrix} 0 & 0 & \ddot{\theta}_1 + \ddot{\theta}_2 \end{bmatrix}^T \quad (A11)$$

The linear velocity of the small connecting rod in the coordinate system {B} is:

$${}^Bv_B = \begin{bmatrix} L_{OB} \cdot \sin\theta_2 \cdot \dot{\theta}_1 & L_{OB} \cdot \cos\theta_2 \cdot \dot{\theta}_1 & 0 \end{bmatrix}^T \quad (A12)$$

The linear acceleration of the small connecting rod in the coordinate system {B} is:

$${}^B\dot{v}_B = \begin{bmatrix} g \cdot \sin(\theta_1 + \theta_2) + L_{OB} \cdot (\sin\theta_2 \cdot \ddot{\theta}_1 - \cos\theta_2 \cdot \dot{\theta}_1^2) \\ g \cdot \cos(\theta_1 + \theta_2) + L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 - \sin\theta_2 \cdot \dot{\theta}_1^2) \\ 0 \end{bmatrix} \quad (A13)$$

The centroid linear acceleration of the small connecting rod in the coordinate system {B} is:

$${}^B\dot{v}_{cB} = \begin{bmatrix} g \cdot \sin(\theta_1 + \theta_2) + L_{OB} \cdot (\sin\theta_2 \cdot \ddot{\theta}_1 - \cos\theta_2 \cdot \dot{\theta}_1^2) - {}^Bp_{cBy} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - {}^Bp_{cBx} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ g \cdot \cos(\theta_1 + \theta_2) + L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 - \sin\theta_2 \cdot \dot{\theta}_1^2) + {}^Bp_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - {}^Bp_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \\ 0 \end{bmatrix} \quad (A14)$$

Then, the resultant force vector Eq. (10) and the resultant torque vector Eq. (11) on the small connecting rod are obtained.

(2) The dynamics are derived inward

According to Eqs. (10) and (12), the combined external force on the small connecting rod is:

$${}^Bf_B = \begin{bmatrix} {}^Bf_{Bx} & {}^Bf_{By} & 0 \end{bmatrix}^T \quad (A15)$$

where,

$${}^Bf_{Bx} = m_2 g \cdot \sin(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\sin\theta_2 \cdot \ddot{\theta}_1 - 2\cos\theta_2 \cdot \dot{\theta}_1^2 - \cos\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) - m_2 \cdot {}^Bp_{cBy} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^Bp_{cBx} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 + F_{Ax}$$

$${}^Bf_{By} = m_2 g \cdot \cos(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 + 2\sin\theta_2 \cdot \dot{\theta}_1^2 + \sin\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) + m_2 \cdot {}^Bp_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^Bp_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2$$

Then, the external torque of the small connecting rod is:

$${}^B\tau_B = \begin{bmatrix} 0 & 0 & {}^B\tau_{Bz} \end{bmatrix}^T \quad (A16)$$

where,

$${}^B\tau_{Bz} = {}^B I_{ZZ} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) + {}^B p_B \cdot [m_2 g \cdot \cos(\theta_1 + \theta_2) + m_2 \cdot L_{OB} (\cos\theta_2 \cdot \ddot{\theta}_1 + 2\sin\theta_2 \cdot \dot{\theta}_1^2 + \sin\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) + m_2 \cdot {}^B p_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^B p_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2]$$

The combined external force on the retral connecting rod is:

$${}^{O_1}f_{O_1} = \begin{bmatrix} {}^{O_1}f_{O_1x} & {}^{O_1}f_{O_1y} & 0 \end{bmatrix}^T \quad (A17)$$

where,

$${}^{O_1}f_{O_1x} = m_2 g \cdot \sin\theta_1 - m_2 \cdot L_{OB} \cdot \theta_1 \cdot (\cos^2\theta_2 + \sin^2\theta_2) \cdot (2\dot{\theta}_1 + \dot{\theta}_2) - m_2 \cdot (\cos\theta_2 \cdot {}^B p_{cBy} + \sin\theta_2 \cdot {}^B p_{cBx}) \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) + 2m_2 \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \cdot (\sin\theta_2 \cdot {}^B p_{cBy} - \cos\theta_2 \cdot {}^B p_{cBx}) + F_{Ax} \cdot \cos\theta_2$$

$${}^{O_1}f_{O_1y} = m_2 g \cdot \cos\theta_1 + 2m_2 \cdot L_{OB} \cdot \ddot{\theta}_1 \cdot (\cos^2\theta_2 + \sin^2\theta_2) + m_2 \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) \cdot (\cos\theta_2 \cdot {}^B p_{cBx} - \sin\theta_2 \cdot {}^B p_{cBy}) - 2m_2 \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \cdot (\sin\theta_2 \cdot {}^B p_{cBx} + \cos\theta_2 \cdot {}^B p_{cBy}) + F_{Ax} \cdot \sin\theta_2$$

Then, the combined torque model of the retral connecting rod, that is Eq. (14), is obtained, that is:

$${}^{O_1}\tau_{O_1} = \begin{bmatrix} 0 & 0 & {}^{O_1}\tau_{O_{1z}} \end{bmatrix}^T \quad (A18)$$

where,

$$\begin{aligned} {}^{O_1}\tau_{O_{1z}} = & {}^B I_{ZZ} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) + {}^B p_{cBx} \cdot [m_2 g \cdot \cos(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 + 2\sin\theta_2 \cdot \dot{\theta}_1^2 + \sin\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) + m^2 \cdot {}^B p_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^B p_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2] + {}^{O_1} I_{ZZ} \cdot \ddot{\theta}_1 \\ & + m_1 g \cdot (\sin\theta_1 \cdot {}^{O_1} p_{cO_{1x}} - \cos\theta_1 \cdot {}^{O_1} p_{cO_{1y}}) - 2m_1 \cdot \ddot{\theta}_1 \cdot {}^{O_1} p_{cO_{1x}} \cdot {}^{O_1} p_{cO_{1y}} - m_1 \cdot \dot{\theta}_1^2 \cdot ({}^{O_1} p_{cO_{1x}}^2 - {}^{O_1} p_{cO_{1y}}^2) + m_2 g \cdot \cos\theta_1 + m_2 \cdot L_{OB} \cdot \ddot{\theta}_1 \cdot (\sin^2\theta_2 + \cos^2\theta_2) + m_2 \cdot (\dot{\theta}_1 \\ & + \ddot{\theta}_2) \cdot (\cos\theta_2 \cdot {}^B p_{cBx} - \sin\theta_2 \cdot {}^B p_{cBy}) - 2m_2 \cdot {}^B p_{cBx} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \cdot (\sin\theta_2 + \cos\theta_2) + F_{Ax} \cdot \sin\theta_2 \end{aligned}$$

According to the same derivation process, the torque model of the mechanism OCD on the joint O, that is Eq. (15), is as follows:

$${}^{O_2}\tau_{O_2} = \begin{bmatrix} 0 & 0 & {}^{O_2}\tau_{O_{2z}} \end{bmatrix}^T \quad (A19)$$

where,

$$\begin{aligned} {}^{O_2}\tau_{O_{2z}} = & {}^C I_{ZZ} \cdot (\ddot{\alpha}_1 + \ddot{\alpha}_2) + {}^C p_{cCx} \cdot [m_3 g \cdot \cos(\alpha_1 + \alpha_2) + m_3 \cdot L_{OC} \cdot (\cos\alpha_2 \cdot \ddot{\alpha}_1 + 2\sin\alpha_2 \cdot \dot{\alpha}_1^2 + \sin\alpha_2 \cdot \dot{\alpha}_1 \cdot \dot{\alpha}_2) + m_3 \cdot {}^C p_{cCx} \cdot (\ddot{\alpha}_1 + \ddot{\alpha}_2) - 2m_3 \cdot {}^C p_{cCy} \cdot (\dot{\alpha}_1 + \dot{\alpha}_2)^2] \\ & + {}^{O_2} I_{ZZ} \cdot \ddot{\alpha}_1 + m_1 g \cdot (\sin\alpha_1 \cdot {}^{O_1} p_{cO_{1x}} - \cos\alpha_1 \cdot {}^{O_1} p_{cO_{1y}}) - 2m_1 \cdot \ddot{\alpha}_1 \cdot {}^{O_1} p_{cO_{1x}} \cdot {}^{O_1} p_{cO_{1y}} - m_1 \cdot \dot{\alpha}_1^2 \cdot ({}^{O_1} p_{cO_{1x}}^2 - {}^{O_1} p_{cO_{1y}}^2) + m_3 g \cdot \cos\alpha_1 + m_3 \cdot L_{OC} \cdot \ddot{\alpha}_1 \cdot (\sin^2\alpha_2 + \cos^2\alpha_2) \\ & + m_3 \cdot (\dot{\alpha}_1 + \dot{\alpha}_2) \cdot (\cos\alpha_2 \cdot {}^C p_{cCx} - \sin\alpha_2 \cdot {}^C p_{cCy}) - 2m_3 \cdot {}^C p_{cCx} \cdot (\dot{\alpha}_1 + \dot{\alpha}_2)^2 \cdot (\sin\alpha_2 + \cos\alpha_2) + F_{Dx} \cdot \sin\alpha_2 \end{aligned}$$

The torque of the mechanism and the mechanism OCD at joint O is equivalent, to establish the dynamic model Eq. (16) of the platen moving phase, that is as follow:

$$\begin{aligned} F_{Dx} = & \left\{ {}^B I_{ZZ} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - {}^C I_{ZZ} \cdot (\ddot{\alpha}_1 + \ddot{\alpha}_2) - {}^C p_{cCx} \cdot [m_3 g \cdot \cos(\alpha_1 + \alpha_2) + m_3 \cdot L_{OC} \cdot (\cos\alpha_2 \cdot \ddot{\alpha}_1 + 2\sin\alpha_2 \cdot \dot{\alpha}_1^2 + \sin\alpha_2 \cdot \dot{\alpha}_1 \cdot \dot{\alpha}_2) + m_3 \cdot {}^C p_{cCx} \cdot (\ddot{\alpha}_1 + \ddot{\alpha}_2) \right. \\ & \left. - 2m_3 \cdot {}^C p_{cCy} \cdot (\dot{\alpha}_1 + \dot{\alpha}_2)^2 \right\} + {}^B p_{cBx} \cdot [m_2 g \cdot \cos(\theta_1 + \theta_2) + m_2 \cdot L_{OB} \cdot (\cos\theta_2 \cdot \ddot{\theta}_1 + 2\sin\theta_2 \cdot \dot{\theta}_1^2 + \sin\theta_2 \cdot \dot{\theta}_1 \cdot \dot{\theta}_2) + m^2 \cdot {}^B p_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) - 2m_2 \cdot {}^B p_{cBy} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2] \\ & + {}^{O_1} I_{ZZ} \cdot \ddot{\theta}_1 - {}^{O_2} I_{ZZ} \cdot \ddot{\alpha}_1 + m_1 g \cdot (\sin\theta_1 - \sin\alpha_1) \cdot {}^{O_1} p_{cO_{1x}} - (\cos\theta_1 - \cos\alpha_1) \cdot {}^{O_1} p_{cO_{1y}} + 2m_1 \cdot {}^{O_1} p_{cO_{1x}} \cdot {}^{O_1} p_{cO_{1y}} \cdot (\ddot{\alpha}_1 - \ddot{\theta}_1) + m_1 \cdot ({}^{O_1} p_{cO_{1x}}^2 - {}^{O_1} p_{cO_{1y}}^2) \cdot (\dot{\alpha}_1^2 - \dot{\theta}_1^2) \\ & + g \cdot (m_2 g \cdot \cos\theta_1 - m_3 g \cdot \cos\alpha_1) - m_3 \cdot L_{OC} \cdot \ddot{\alpha}_1 \cdot (\sin^2\alpha_2 + \cos^2\alpha_2) + m_2 \cdot L_{OB} \cdot \ddot{\theta}_1 \cdot (\sin^2\theta_2 + \cos^2\theta_2) - m_3 \cdot \cos\alpha_2 \cdot {}^C p_{cCx} \cdot (\ddot{\alpha}_1 + \ddot{\alpha}_2) + m^2 \cdot \cos\theta_2 \cdot {}^B p_{cBx} \cdot (\ddot{\theta}_1 + \ddot{\theta}_2) \\ & + 2m_3 \cdot {}^C p_{cCy} \cdot (\dot{\alpha}_1 + \dot{\alpha}_2)^2 \cdot (\sin\alpha_2 + \cos\alpha_2) - 2m_2 \cdot {}^B p_{cBx} \cdot (\dot{\theta}_1 + \dot{\theta}_2)^2 \cdot (\sin\theta_2 + \cos\theta_2) + F_{Ax} \sin\theta_2 \} / \sin\alpha_2 \end{aligned} \quad (A20)$$

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